

Git 4

A **merge conflict** occurs when Git cannot automatically merge changes because the same part of a file has been modified differently in two branches.

Attempt to merge the branches

```
git merge feature
```

If there is a conflict, Git will stop the merge and display a conflict message.

Identify the conflicted files

```
git status
```

Git lists the files with merge conflicts.

Open the conflicted file

You will see conflict markers like:

```
<<<<<< HEAD
```

```
Hello from Main Branch
```

```
=====
```

```
Hello from Feature Branch
```

```
>>>>>> feature
```

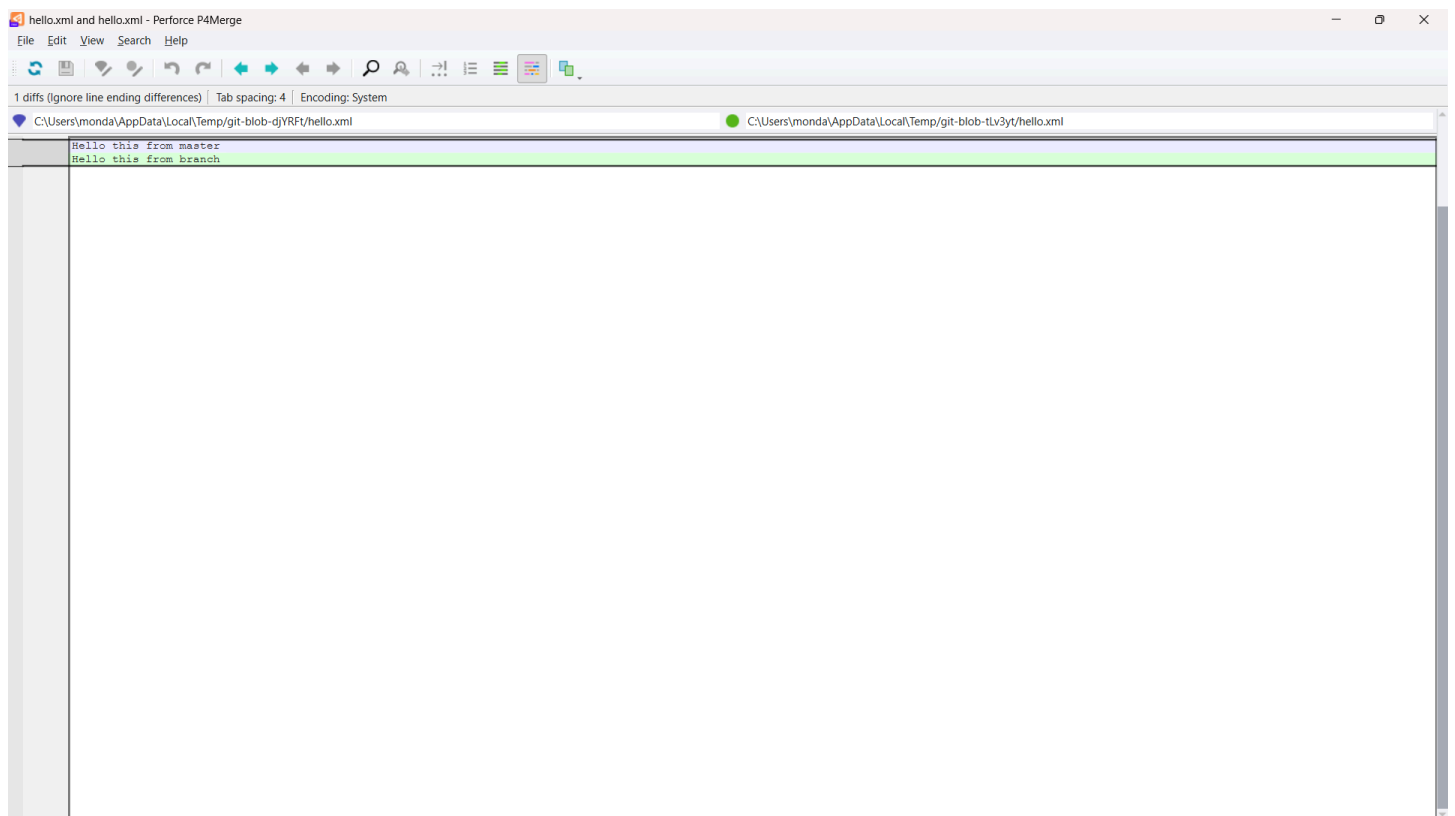
- HEAD shows the current branch's changes.
- feature shows the incoming branch's changes.

Resolve the conflict

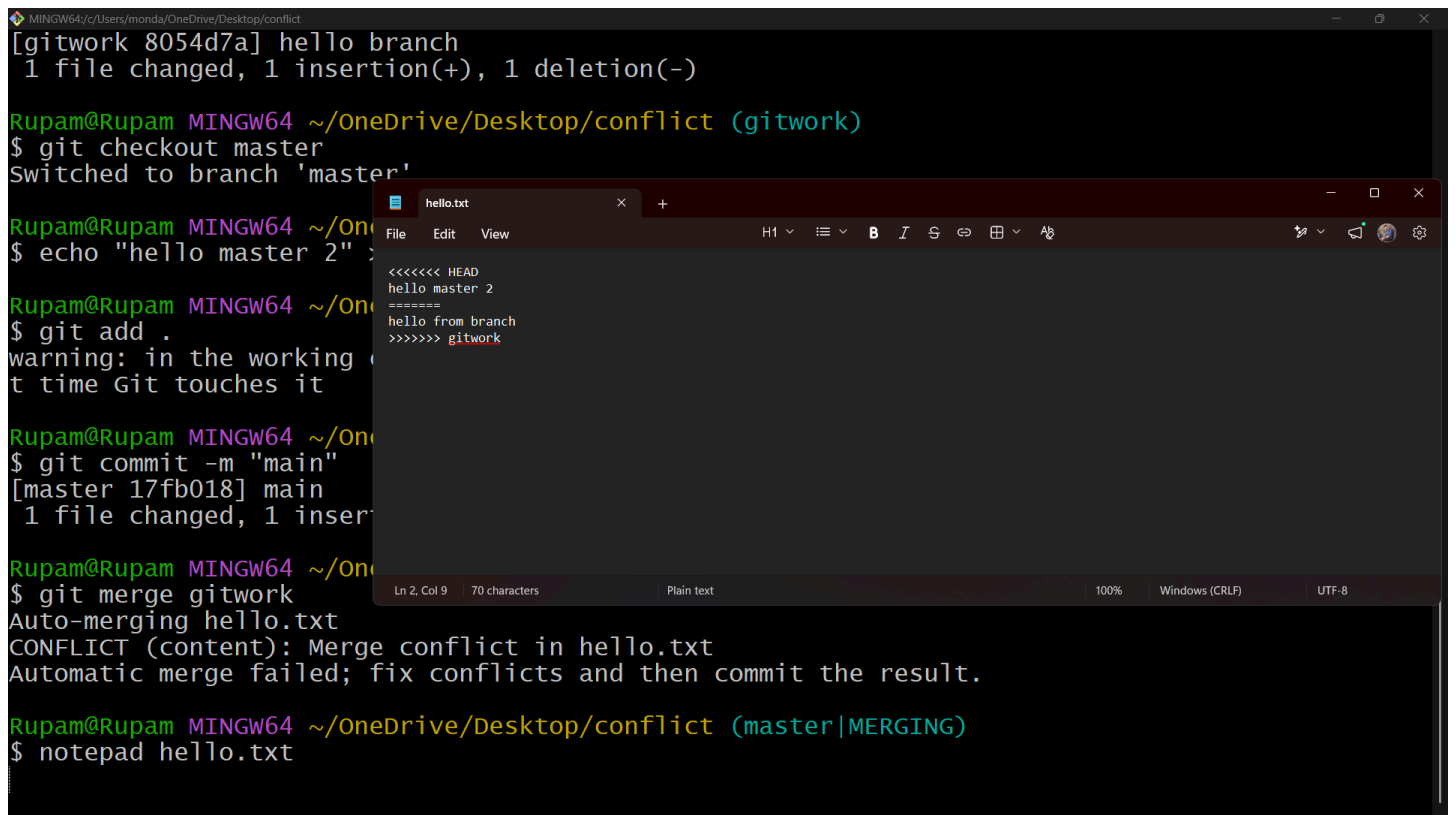
Edit the file by choosing one version, combining both versions, or writing new content.

Remove the conflict markers (<<<<<<, =====, >>>>>>).

p4merge tool



GitBash



```
MINGW64/c/Users/monda/OneDrive/Desktop/conflict
1 file changed, 1 insertion(+), 1 deletion(-)

Rupam@Rupam MINGW64 ~/OneDrive/Desktop/conflict (master)
$ git merge gitwork
Auto-merging hello.txt
CONFLICT (content): Merge conflict in hello.txt
Automatic merge failed; fix conflicts and then commit the result.

Rupam@Rupam MINGW64 ~/OneDrive/Desktop/conflict (master|MERGING)
$ notepad hello.txt

Rupam@Rupam MINGW64 ~/OneDrive/Desktop/conflict (master|MERGING)
$ git add .

Rupam@Rupam MINGW64 ~/OneDrive/Desktop/conflict (master|MERGING)
$ git commit -m "final"
[master 77817d6] final

Rupam@Rupam MINGW64 ~/OneDrive/Desktop/conflict (master)
$ git branch
gitwork
* master

Rupam@Rupam MINGW64 ~/OneDrive/Desktop/conflict (master)
$ git branch -d gitwork
Deleted branch gitwork (was 8054d7a).

Rupam@Rupam MINGW64 ~/OneDrive/Desktop/conflict (master)
$
```

```
MINGW64/c/Users/monda/OneDrive/Desktop/conflict

Rupam@Rupam MINGW64 ~/OneDrive/Desktop/conflict (master|MERGING)
$ notepad hello.txt

Rupam@Rupam MINGW64 ~/OneDrive/Desktop/conflict (master|MERGING)
$ git add .

Rupam@Rupam MINGW64 ~/OneDrive/Desktop/conflict (master|MERGING)
$ git commit -m "final"
[master 77817d6] final

Rupam@Rupam MINGW64 ~/OneDrive/Desktop/conflict (master)
$ git branch
gitwork
* master

Rupam@Rupam MINGW64 ~/OneDrive/Desktop/conflict (master)
$ git branch -d gitwork
Deleted branch gitwork (was 8054d7a).

Rupam@Rupam MINGW64 ~/OneDrive/Desktop/conflict (master)
$ git log --oneline --decorate
77817d6 (HEAD -> master) final
17fb018 main
8054d7a hello branch
9cff8c3 master

Rupam@Rupam MINGW64 ~/OneDrive/Desktop/conflict (master)
$
```